

# TRELLIS.2 Printability Fine-Tuning - Weekly Report

## Abstract

Many post-processing libraries and methods were tried to improve quality and printability, adding different options for post-processing. manifold3d was found to be the best for 3D printing, and the original trimesh pipeline was kept as well. Further research was done on the best way to decrease non-manifold edges of the models, and the TriFlow paper on generating artist-like 3D models was read.

Integrating DreamDPO concepts was also tried, with no success. The DreamDPO work for diffusion models was read, but the preference discriminator was simulated as a score rather than a VLM, and the results showed no gains - it always chose the reference path for the flow model. That, together with a lack of expertise in this area of the theory, led to the decision to work instead on fine-tuning the SLat flow model using LoRA methods.

This was tried on a small sample size of 300 curated 3D models from Thingi10K. Initial training showed gains, with significantly less non-manifold edges for raw results, without loss in detail. More latent data is being collected for training.

## 1. Executive summary

We retired the inference-time DPO steering approach, established that supervised fine-tuning is the better instrument for this problem, built the full data and training pipeline, and completed a 1,170-step LoRA fine-tune of TRELLIS.2's shape-SLat flow model on 300 clean Thingi10K models.

The fine-tune works on the axes it can reach. Measured on exported meshes, held-out set, paired sampling noise:

| Model      | L_OH (overhang) | L_Th (thin walls) | L_Topo        | R_Detail     |
|------------|-----------------|-------------------|---------------|--------------|
| Base model | 0.2291          | 0.1133            | 0.1871        | 0.4316       |
| step 750   | 0.2183 (-5%)    | 0.0152 (-87%)     | 0.0500 (-73%) | 0.4578 (+6%) |
| step 900   | 0.2255 (-2%)    | 0.0299 (-74%)     | 0.0635 (-66%) | 0.4411 (+2%) |
| step 1050  | 0.2179 (-5%)    | 0.0159 (-86%)     | 0.0340 (-82%) | 0.4358 (+1%) |

Step 1050 beats the base model on thin-wall penalty for 6 of 6 held-out objects. Image fidelity did not degrade (0.897 -> 0.908-0.913), and surface detail slightly increased, so this is not printability bought by smoothing the model into generic blobs.

## 2. What was discovered

### 2.1 The retired approach failed for a diagnosable reason

Inference-time DPO steering produced a gradient contribution of -0.0027 +/- 0.0078 over 17 scored forks - indistinguishable from zero. Root cause: the only differentiable scalar available (slat\_detail\_proxy) agreed with the judge's within-fork argmax 5.9% of the time against a 33% chance line, with mean Spearman -0.265. The optimiser worked; its objective was near-orthogonal to the goal.

### 2.2 The "differentiability wall" did not exist

### 2.3 The VAE bounds what any flow-model training can achieve

| Property                | Result                                                               |
|-------------------------|----------------------------------------------------------------------|
| Shape fidelity          | Preserved - Chamfer 0.0052, p95 0.0101, bounding box exact           |
| Watertightness          | 0 of 2 preserved                                                     |
| Defect rate (full mesh) | 0.086% open edges, 0.095% non-manifold                               |
| Components              | 953, but largest is 99.85% of faces - and still not watertight alone |

Consequence: geometry-carried printability (thickness, overhang, bulk shape) is reachable by

Rather than  
gains, and  
thing from

The premises  
regresses  
only connect  
inspection

Encoding a

training the flow model; watertightness is not, and belongs to post-processing. This single 20-minute measurement redirected the entire project.

## 2.4 The judge inflates topology defects ~50x

geometric\_  
rate on the  
L\_Topo fig  
defects. An

## 2.5 Measurement noise nearly made the whole run uninterpretable

Two evalua  
sampling m  
held-out ob  
bit-identica

## 2.6 Silent failures dominated the engineering cost

Every sign

| Symptom                         | Actual cause                                        | Cost                   |
|---------------------------------|-----------------------------------------------------|------------------------|
| Training step took 464s         | No activation checkpointing -> 38 GB retained       | 18x slowdown           |
| Baseline eval returned n_eval=0 | Three stacked bugs, masked by except: contin        | 2 invalidated launches |
| Eval decoder on CPU             | low_vram is device placement, not a memory hi       | 1 launch               |
| Sample-evolution panel empty    | A patch silently failed to match; the call site was | whole run, cosmetic    |
| Loss chart nonsense             | Dataset build logged token counts into the loss f   | misleading chart       |

The single highest-leverage change all week was replacing except Exception: continue with recorded, printed failures. It converted two blind retries into one-step diagnoses.

## 2.7 Post-processing was addressed before generation was

Many post-  
different op  
was added  
best way to  
was read.

This matters for reading the rest of this report: the fine-tuning results quoted here are on raw decoder output, with no post-processing applied. The reduction in non-manifold edges is therefore a gain made before manifold3d or trimesh repair runs, not instead of them. Both remain in the pipeline, and the round-trip result in sec.2.3 is why - watertightness is a decoder property that fine-tuning cannot reach.

## 3. What was built

| Component                | File                           | Note                                          |
|--------------------------|--------------------------------|-----------------------------------------------|
| CPU mesh -> SLat encoder | backends/dual_grid_cpu.py      | o-voxel ships CUDA-only; rebuilt the pure-C++ |
| Offscreen renderer       | trellis_core/render_mesh.py    | moderngl, Metal-backed, headless; camera fitt |
| Dataset pipeline         | trellis_core/dataset_build.py  | 300 models, 3.2 GPU-hours, 0 failures         |
| LoRA                     | trellis_core/lora.py           | 210 modules, 18.68M params (1.4% of 1.29B)    |
| SFT trainer              | trellis_core/sft_train.py      | Hyperparameters from TRELLIS.2's own config   |
| Live dashboard           | devlab/train_server.py         | Crash-independent; phone access over Tailsca  |
| VAE round-trip test      | trellis_core/vae_roundtrip.py  | Established the ceiling                       |
| Mesh exporter            | trellis_core/export_samples.py | Reproduces eval exactly (same seeds, same     |
| Topology test            | trellis_core/topology_test.py  | Paired, undecimated, multi-seed               |

## Dataset

Thingi10K  
3D-printing  
multi-comp  
PWN + no  
(270 train /

## Training run

```
steps      1,170 of 1,200 (stopped after divergence)
epochs     8.7 over 270 examples
GPU time   9.96 hours, mean 30.7 s/step
LoRA       rank 16, alpha 32, lr 1e-4 constant, AdamW wd 0.01
schedule   uniform t, p_uncond 0.1 (both from TRELLIS.2's official config)
evals      10, every 150 steps, 6 held-out objects, common random numbers
```

run id

## 4. The divergence

| Steps     | Mean loss | Max loss | Grad norm |
|-----------|-----------|----------|-----------|
| 1-1100    | ~0.16     | 0.80     | 0.03-0.06 |
| 1101-1170 | 3.48      | 57.29    | 332.25    |

Held-out loss went 0.0859 -> 10.29; image similarity 0.908 -> 0.202. The model was destroyed.

Two causes, both mine: no learning-rate decay (constant 1e-4 for 8.7 epochs), and gradient clipping that bounded each update but could not prevent AdamW's moments accumulating through a sustained blow-up. Checkpoints 750/900/1050 were preserved before stopping, because a clean stop writes a final checkpoint and `prune_checkpoints(keep=3)` would have deleted step 750.

## 5. How to log improvements numerically

Recommended standard for every future comparison. The first three are non-negotiable given what we measured this week.

### 5.1 Always paired, never pooled

Fix the sam  
per-object  
is between  
within-fork

### 5.2 Always against a baseline

Every run h  
number wi

### 5.3 Always report fidelity beside printability

Minimum p  
conditioni  
experimen

### 5.4 The metric set

|                             |                       |         |                                         | Metric |
|-----------------------------|-----------------------|---------|-----------------------------------------|--------|
| L_Th                        | judge                 | lower   | Responds strongly. Primary succe        |        |
| nonmanifold_rate, open_rate | full mesh             | lower   | Must be computed undecimated (          |        |
| L_OH                        | judge                 | lower   | Has not responded; also only ~5%        |        |
| R_Detail                    | judge                 | context | Fidelity term, not printability - no be |        |
| image_similarity            | silhouette IoU        | higher  | Prompt-adherence guard                  |        |
| sample_dispersion           | latent means          | flat    | Collapse guard                          |        |
| watertight_rate             | full mesh             | control | Expected flat; a change indicates a     |        |
| support volume              | slicer, not yet built | lower   | The only metric outside the trainin     |        |

## 5.5 The acceptance bar

$$\text{capture} = (\text{tuned\_1\_sample} - \text{base\_1\_sample}) / (\text{base\_best\_of\_N} - \text{base\_1\_sample})$$

A single sample from the tuned model must recover a meaningful fraction of what N-fold rejection sampling buys from the base model at matched compute. This test has not yet been run.

## 6. Next steps

1. Finish the topology consistency test (running: 10 objects x 3 seeds x 4 models, paired, undecimated). Answers whether step 1050's cleaner topology is consistent or a lucky draw.
2. Run the best-of-N comparison (sec.5.5). Until this exists we do not know whether the fine-tune beats cheap rejection sampling - the only comparison that justifies the work.
3. Fix the training schedule and re-run. Cosine LR decay; a skip-step rule that drops a batch whose gradient norm exceeds a multiple of the running median rather than clipping it; keep every checkpoint. The metrics were still improving when the run broke, so the ceiling was never observed.
4. Repair the judge before it is trusted further: make the thickness raycast deterministic (test-retest reliability was measured at 0.208, capping any correlation at  $|r| \leq 0.456$ ), and add a support raycast to the overhang term (currently rank-correlates -0.432 with true unsupported area).
5. Scale the dataset. 270 training objects is the binding constraint, not step count; 3,935 clean models are available and prep runs at ~40 s/model.
6. Add a slicer-based support-volume metric as the held-out physical measure that was never part of the training reward.

## 7. Deliverables on disk

|                                          |                                          |
|------------------------------------------|------------------------------------------|
| runs/sft-1200-20260806-0119/             |                                          |
| ckpt_keep/step_{750,900,1050}/adapter.pt | preserved LoRA adapters (643 MB)         |
| meshes/step_{0,750,900,1050}/            | 24 GLB + 24 STL, raw, no post-processing |
| exports/                                 | 42 PNG frames + 6 progression strips     |
| metrics.jsonl                            | 1,170 train + 10 eval records            |
| topology_test.jsonl                      | in progress                              |
| data/thingi10k_sft/                      | 300 latents + images + conditioning      |

All meshes are raw decoder output - no repair, no decimation, no solidify. Each object uses identical conditioning and identical sampling noise across all four models, so differences are attributable to the model alone.